Solutions

1. Solution to Question on Wind and Photovoltaic Power

a)

Power through rotor circuit $P_r = sP_s$

$$\frac{P_r}{P_s + P_r} = 0.2 \Rightarrow \frac{s}{s+1} = 0.2 \Rightarrow s = 0.25$$

[2 marks]

Ratio of maximum to minimum rotor speed

$$\frac{1+s}{1-s} = \frac{5}{3} = 1.667$$

[1 mark]

b)

i)

$$P_w = \frac{1}{2}\rho A v_w^3 C_{p-max}$$

$$\Rightarrow v_R = \left(\frac{2P_{rated}}{\rho\pi r^2 C_{p-max}}\right)^{\frac{1}{3}} = 11.158 \text{ m/s}$$

[1 mark]

$$p = \Phi(11.158) = 81.3\%$$

[1 mark]

ii)

$$p = 1 - [\Phi(26) - \Phi(3)] = 12.93\%$$

[1 mark]

iii)

$$p = 1 - [\Phi(26) - \Phi(11.158)] = 18.67\%$$

[1 mark]

iv)

$$\text{TSR } \lambda = \frac{\omega_T \times r}{v_w} \Rightarrow \omega_{T-opt} = \frac{\lambda_{opt} \times v_w}{r}$$

$$\left(\frac{\lambda_{opt} \times v_{w-cutin}}{r} \times \frac{60}{2\pi}\right) \leq \omega_{T-opt} \leq \left(\frac{\lambda_{opt} \times v_R}{r} \times \frac{60}{2\pi}\right)$$

Model Answers

$\Rightarrow 3.19 \text{ rpm} \leq \omega_{T-opt} \leq 11.87 \text{ rpm}$

[1 mark]

$$\frac{1+s}{1-s} = \frac{11.87}{3.19} \Rightarrow s = 0.5762$$

[1 mark]

$$\frac{P_{conv}}{P_{Total}} = \frac{sP_s}{P_s + sP_s} = 36.56\%$$

[1 mark]

c)

i)

$$0 = 4.1 - I_0\left[e^{\frac{qV_{oc}}{kT}} - 1\right]$$
$$\Rightarrow V_{oc} = \frac{kT}{q}\ln\left(\frac{4.1 + I_0}{I_0}\right)$$

[1 mark]

$V_{oc} = 0.8023$ V

[1 mark]

ii)

$$I_{SC} = 4.1 - I_0\left[e^{\frac{q\times 0}{kT}} - 1\right] = 4.1 \text{ A}$$

[1 mark]

iii)

$R = 0.1 = \frac{V}{I}$
$V = 4.1 \times 0.1 = 0.41$ V
$P = V \times I = 1.681$ W

[1 mark]

iv)

$$R = 1 = \frac{V}{I}$$
$I \approx V_{oc} \times 1 = 0.0.8023$ A
$P \approx V_{oc} \times I = 0.6437$ W
Less power would be drawn by the 1 Ω resistor

[1 mark]

Model Answers

d)

i)

$$\eta = \frac{P_r}{1000 \times A} = \frac{350}{1000 \times 1.35} = 25.93\%$$

[2 marks]

ii)

$$P_{750} = N_{mod} \times 750 \times A \times \eta = 262.5 \text{ kW}$$

[1 mark]

iii)

The current fed by the affected strings would be zero as the corresponding string diodes are reverse biased. Hence power output under STC will be

$$P_{\text{out}} = N_{\text{mod}} \times 350 \times \frac{40}{50} = 280 \text{ kW}$$

[1 mark]

iv)

The current fed by one affected string with 10 shaded PV modules would be zero. Hence power output under STC will be

$$P_{\text{out}} = N_{\text{mod}} \times 350 \times \frac{49}{50} = 343 \text{ kW}$$

[1 mark]

Model Answers

2. Solution to Question on balancing, Voltage Control and Security of Supply

Part A:

a) Statements along the following lines would be sufficient for the full mark:

- Part-loaded generation, including
  - Enhancing flexibility of nuclear generation
- Pumped-storage hydro generation
- Flexibility
- Standby generation (standing reserve)

b) Statements along the following lines would be sufficient for the full marks:

- Advantages
  - Cost effective solution in providing reserve
  - Increased efficiency of generation that is not anymore part loaded
  - Potentially less curtailment of renewables

- Disadvantages
  - More actors involved increasing complexity and adds need for coordination
  - Load recovery for some of technologies, such as DSR, Energy Storage, V2G
  - Might need contracting more capacity due to potentially higher uncertainty of some of technologies

Part B:

c)

The answer along the following lines will give full marks.

Factors:

- Circuit capacity
- Voltage constraints
- Fault level constraints
- Voltage distortion/harmonics (for long connections)

Drivers:

- Connection voltage
- Impedance of connecting circuit (including length of connection circuit(s)) that affect voltage rise (including voltage drop on adjacent feeders)
- Peak and minimum local demand
- Reactive power (including reactive compensation)

[1 mark; 0.25 marks for each relevant pair factor/driver with the maximum of 1 mark in total]

d)

-Wind / solar curtailment: during critical conditions wind generation is curtailed to keep network constraints within limits.
-Reactive power compensation: reactive compensator at wind / solar farm connection point could be set or controlled in a such way to reduce the impact of voltage rise effect.
-Voltage control: different voltage control techniques (setting different tap changer position/target voltage, adding in-line voltage regulators and area-based voltage control) could be applied to keep voltage withing limits.

Model Answers

[3 marks; 1 mark for each relevant approach with the maximum of 3 marks in total]

e)

Capacity outage probability table for the generation plant is

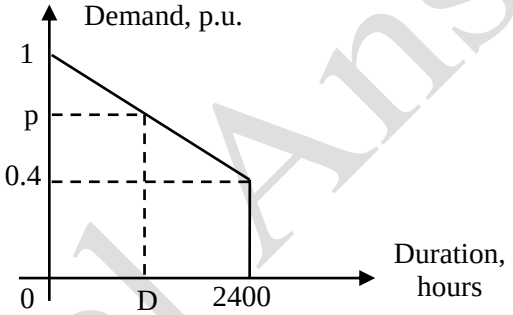
Capacity in service, MW	Probability
60	0.9 x 0.8 x 0.6 = 0.432
50	0.9 x 0.8 x (1-0.6) = 0.288
40	0.9 x (1-0.8) x 0.6 = 0.108
30	0.9 x (1-0.8) x (1-0.6) + (1-0.9) x 0.8 x 0.6 = 0.12
20	(1-0.9) x 0.8 x (1-0.6) = 0.032
10	(1-0.9) x (1-0.8) x 0.6 = 0.012
0	(1-0.9) x (1-0.8) x (1-0.6) = 0.008

The total probability for all capacity states is 1.

[4 marks; subtract 1 mark for any missing capacity in service and wrong probability up to a maximum of subtracted 4 marks]

f)

Energy not supplied is calculated from the LDC in the following way



For a given $p \geq 0.4$, D could be calculated from linear equation:

$$D = \frac{2400 - 0}{0.4 - 1}(p - 1) = 4000 \cdot (1 - p)$$

And energy not supplied, ENS, is:

$$ENS = 0.5 \cdot (1 - p) \cdot C \cdot D$$

Where total capacity, C, is 60 MW.

For a given $p \leq 0.4$, ENS could be calculated as

$$ENS = 0.5 \cdot (1 - 0.4) \cdot C \cdot 2400 + (0.4 - p) \cdot C \cdot 2400 = (0.7 - p) \cdot C \cdot 2400$$

Expected energy not supplied is equal to $ENS \cdot Probability$

Capacity, MW	Probability	p	D, hours	ENS, MWh	EENS, MWh
60	43.2%	1	0	0	0
50	28.8%	0.833333	666.6667	3333.333	960
40	10.8%	0.666667	1333.333	13333.33	1440

Model Answers

30	12.0%	0.5	2000	30000	3600
20	3.2%	0.333333		52800	1689.6
10	1.2%	0.166667		76800	921.6
0	0.8%	0		100800	806.4
Total	**100.0%**				**9417.6**

Total EENS is 9417.6 MWh.

[5 marks; subtract 2 marks if it not clearly explained how EENS is calculated; subtract 1 mark for any wrong pair ENS/EENS, including total EENS up to a maximum of subtracted 5 marks]

g)

Energy not supplied when $p = 0.4$ is

$$ENS_{0.4} = 0.5 \cdot (1 - 0.4) \cdot 60 \cdot 2400 = 43{,}200MWh$$

Given the equivalent firm capacity source never fails, probability is 1 and hence firm capacity is greater than 0.4 p.u.

$$ENS = 0.5 \cdot (1 - p) \cdot C \cdot D = 0.5 \cdot (1 - p) \cdot C \cdot 4000 \cdot (1 - p)$$

$$(1 - p)^2 = \frac{ENS}{120000}$$

$$p = 1 - \sqrt{\frac{ENS}{120000}} = 1 - \sqrt{\frac{9417.6}{120000}} = 0.720$$

Contribution is 72% and in MW is $0.72 \cdot 60 = 43.2MW$. Firm capacity is 43.2MW.

[3 marks; subtract 2 marks if it is not clearly explained how p *or* $p \cdot C$ is calculated; subtract 1 mark for wrong firm capacity]

Model Answers

3. Solution to Question on Low Carbon Heat

a)

i) The purpose of the CCC is to:

- advise UK Government and Devolved Administrations on emissions targets and report to Parliament on progress made in reducing greenhouse gas emissions and preparing for climate change.
- provide independent advice on setting and meeting carbon budgets and preparing for climate change
- monitor progress in reducing emissions and achieving carbon budgets and targets
- conduct independent analysis into climate change science, economics and policy
- engage with a wide range of organisations and individuals to share evidence and analysis

[2 marks]

ii) The UK Government should:

- legislate for Net Zero greenhouse gas (GHG) by 2050 along with plans and policies
- set a target for 2033-37 of 965Mt CO_{2eq}
- include all GHG emissions including international aviation, shipping and CO_2 removals
- be based on actual UK emissions without recourse to international credits

[2 marks]

b)

Building space heat demand (W_{th})
= (Tint – T ext) * Average thermal transmittance * Building floor area
= (20-7) *3*100 = **$\underline{3900W_{th}}$**

Where:
Tint = Building internal set temperature, °C
Text= Building external set temperature, °C

Assumptions:
Estimate of heat demand does not take account of:

- other weather factors such as wind, solar gain.
- building air changes
- other heat sources

[3 marks]

c)

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle, and these comprise:

- Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.

Model Answers

- Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.
- Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
- Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.

[4 marks]

d)

i) Building space heat demand (W_{th})
= (Tint – T ext) * Average thermal transmittance * Building floor area
= (20-(-5)) *3*100 = **7500W_{th}**

[3 marks]

ii) If the householder selects the 5kW_{th} ASHP, then re-arranging the above equation
Tint = (5000/ (3*100))-5= **11.7°C**

[3 marks]

iii) Re-arranging the above equation:
Average thermal transmittance
= ASHP, W_{th} / (building floor area*(Tint – T ext))
=5000/ (100*(20-(-5))
=2W/m^2K

Hence improvement in average building thermal transmittance = (1-2/3) = **33%**

[3 marks]

Model Answers

4. Solution to Question on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a) Key similarity between DSR and energy storage is that both can effectively shift demand in time to enable more cos-efficient system operation and investment. This is achieved by reducing electricity consumption during periods of shortage or high cost and increasing demand during periods of high supply availability and low system demand. Both DSR and energy storage can provide up and down regulation i.e. effectively increase or reduce power consumption at different points in time, but unlike generators that only inject power into the grid, have to offset any demand reduction with demand increase at another time.
[0.5 marks for the explanation of shifting demand in time]

One disadvantage of energy storage is that it typically incurs cycle of roundtrip losses, which means some of the energy stored is lost in the process of charging and discharging the storage assets. DSR typically does not have this issue when it shifts demand in time.

[0.5 marks for the explanation of storage cycle losses]

DSR on the other hand is more constrained in its operation because of customer-level constraints i.e. because of the need to deliver useful services to customers without compromising on their comfort level (e.g. EVs still need to be charged by a certain time, heating energy delivered to the house or washing machine cycle completed by the morning). Energy storage on the other hand is not constrained in this way and can maximise its utilisation to support the system.

[0.5 marks for the explanation of customer-level constraints for DSR]

Finally, energy storage requires investment in new energy assets to be able to support system operation. DSR typically uses flexible loads that are already there, with the only additional investment required to enable appropriate ICT/smart control functionality to support efficient demand shifting.

[0.5 mark for the explanation of asset investment cost of storage]

b)

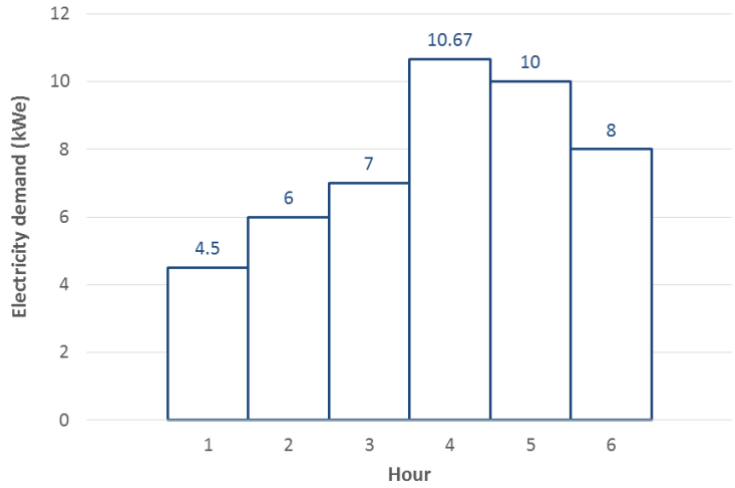
i) Peak electricity demand is found as the highest ratio between heat demand and COP across all hours.

Peak electricity demand = 16.0 / 1.5 = **10.67 kW_{el}**.

[1 mark]

The electricity demand diagram is shown below.

Model Answers



[1 mark]

ii) We start the calculation on the hypothesis (which we will later verify) that TES is only required during the 2 hours with the highest demand, i.e. Hours 4 and 5. Electricity demand during these two hours should ideally be reduced to the same level if allowed by TES parameters. Let x denote the utilisation / thermal power of TES released during Hour 4 that is needed to reduce electricity demand in Hours 4 and 5 to the same level. The following then has to hold:

$(16\ kW_{th} - x) / 1.5 = (15\ kW_{th} - (7\ kWh_{th} - x)) / 1.5$

Solving this for x gives: $x = 4\ kW_{th}$, which does not exceed the TES nominal heat power and therefore is a feasible solution for Hour 4. The optimal utilisation of TES during Hours 4 and 5 is therefore:

- Hour 4: 4 kW_{th}
- Hour 5: 3 kW_{th}

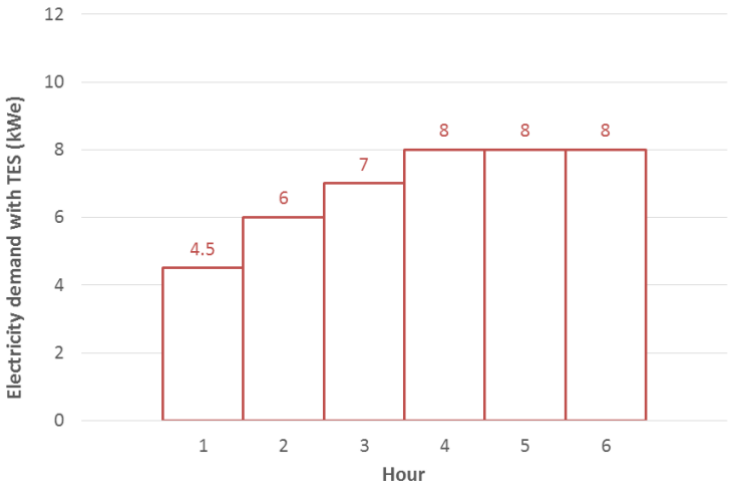
[2 marks for the correct utilisation schedule of TES]

The peak electricity demand now occurs in Hours 4 and 5 (as well as Hour 6), and is equal to 12 kW_{th} / 1.5 = **8 kW_{el}**, i.e. the peak electricity demand is **reduced by 2.67 kW_{el}** compared to the case without TES. It can also be verified that no other hours exceed this level of peak demand (in which case we would need to recalculate the utilisation of storage and peak demand reduction to consider those additional hours).

[2 marks for the correct level of peak demand reduction]

The electricity demand diagram after applying TES is shown below.

Model Answers



[2 marks for the correct chart]

Hydro Power

c)

i)

$$P = \rho g Q H \eta$$

$$\Rightarrow H = \frac{P}{\rho g Q \eta} = 110.8 \text{ m}$$

[1 mark]

$$h_{hw} = h_{tw} + H = 130.8 \text{ m}$$

[1 mark]

ii)

Cross-sectional area of the penstock

$$A_p = \pi R^2 = 78.54 \text{ m}^2$$

Velocity of water at penstock inlet

$$v_2 = \frac{Q}{A_p} = 12.73 \text{ m/s}$$

[1 mark]

Using Bernoulli's principle

$$p_{atm} + \rho g h_1 = p_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2$$

[1 mark]

Model Answers

$$\Rightarrow p_2 = p_{atm} + \rho g(h_1 - h_2) - \frac{1}{2}\rho v_2^2 = 812.92 \text{ kPa}$$

[1 mark]

Offshore Transmission

d)

i)

- The charging current drawn by the 250 mile high voltage cable is significant compared to current carrying capacity of the conductor. Hence, HVDC is necessary instead of AC.
- There are strong AC systems at both ends (main grids in Scotland and England & Wales) to allow LCC-HVDC to operate satisfactorily.
- The converter stations at both ends are located on land. So large substation footprint required for LCC-HVDC is not a problem.
- VSC-HVDC could have been used but was not available with 2.2 GW rating at the time this project was planned.

[1 mark each for any two points - 2marks]

ii)

- Charging current drawn by a 120 km long cable would normally necessitate HVDC connection.
- For Hornsea One wind farm, the shallow seabed on the way to UK shoreline meant installation of an intermediate reactive compensation substation was more cost effective than using HVDC.
- Hence Hornsea One used AC connection to the onshore grid.

[1 mark each – 2 marks]

iii)

- Charging current drawn by a 100 km long cable would necessitates HVDC connection.
- This has to be VSC HVDC due to its less reliance on AC system strength, use of stronger and lighter polymeric cables and compact HVDC converter station which can be fitted on to an offshore platform.

[0.5 mark each]

Model Answers